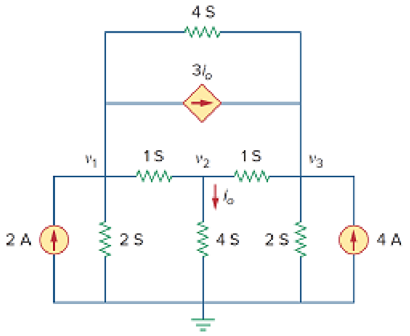
Chapter 3, Problem 27P

Problem

Use nodal analysis to determine voltages $v1$, $v2$, and $v3$ in the circuit of Fig. 3.76.

Figure 3.76:



Step-by-step solution

Step 1 of 10

Refer to Figure 3.76 in the textbook.

Apply nodal analysis at node v_1.

From Figure 3.76,

$i_o = 4v_2$ …… (1)

$$-2+\frac{v_1}{(1/2)}+\frac{v_1-v_2}{(1/1)}+3i_o+\frac{v_1-v_3}{(1/4)}=0$$

$2v_1+v_1-v_2+3i_o+4v_1-4v_3=2$ …… (2)

Step 2 of 10

Replace equation 2 in equation 1.

$$2v_1+v_1-v_2+3(4v_2)+4v_1-4v_3=2$$
$$2v_1+v_1-v_2+12v_2+4v_1-4v_3=2$$

$7v_1+11v_2-4v_3=2$ …… (3)

Apply nodal analysis at node 2.

$$1(v_2-v_1)+4v_2+v_2-v_3=0$$

$6v_2-v_1-v_3=0$ …… (4)

Step 3 of 10

Apply nodal analysis at node 3.

$$1(v_3-v_2)+4(v_3-v_1)-3i_o+2v_3-4=0$$
$$7v_3-v_2-4v_1-3(4v_2)-4=0$$
$$7v_3-v_2-4v_1-12v_2-4=0$$

$7v_3-4v_1-13v_2=4$ …… (5)

Step 4 of 10

Consider the matrix as,

$$\begin{bmatrix} 7 & 11 & -4 \\ -1 & 6 & -1 \\ -4 & -13 & 7 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}$$

Determine the value of Δ.

$$\begin{aligned}\Delta &= \left(7(42-13)-11(-7-4)-4(13+24)\right) \\ &= 176\end{aligned}$$

Step 5 of 10

Determine the value of Δ_1.

Consider the matrix as,

$$\begin{bmatrix} 2 & 11 & -4 \\ 0 & 6 & -1 \\ 4 & -13 & 7 \end{bmatrix}$$

$$\begin{aligned}\Delta_1 &= \left(2(42-13)-11(4)-4(-24)\right) \\ &= 110\end{aligned}$$

Step 6 of 10

Determine the value of Δ_2.

Consider the matrix as,

$$\begin{bmatrix} 7 & 2 & -4 \\ -1 & 0 & -1 \\ -4 & 4 & 7 \end{bmatrix}$$

$$\begin{aligned}\Delta_2 &= \left(7(4)-2(-7-4)-4(-4)\right) \\ &= 66\end{aligned}$$

Step 7 of 10

Determine the value of Δ_3.

Consider the matrix as,

$$\begin{bmatrix} 7 & 11 & 2 \\ -1 & 6 & 0 \\ -4 & -13 & 4 \end{bmatrix}$$

$$\begin{aligned}\Delta_3 &= \left(7(24)-11(-4)+2(13+24)\right) \\ &= 286\end{aligned}$$

Step 8 of 10

Determine the value of the voltage v_1.

Consider the equation for the v_1.

$$v_1 = \frac{\Delta_1}{\Delta} \quad \text{...... (6)}$$

Replace 110 for Δ_1, and 176 for Δ in equation 6.

$$v_1 = \frac{110}{176}$$
$$= 0.625 \text{ V}$$

Therefore, the value of the voltage v_1 is $\boxed{625 \text{ mV}}$.

Step 9 of 10

Determine the value of the voltage v_2.

Consider the equation for the v_2.

$$v_2 = \frac{\Delta_2}{\Delta} \quad \text{...... (7)}$$

Replace 66 for Δ_2, and 176 for Δ in equation 7.

$$v_2 = \frac{66}{176}$$
$$= 0.375 \text{ V}$$

Therefore, the value of the voltage v_1 is $\boxed{375 \text{ mV}}$.

Step 10 of 10

Determine the value of the voltage v_3.

Consider the equation for the v_3.

$$v_3 = \frac{\Delta_3}{\Delta} \quad \text{...... (8)}$$

Replace 286 for Δ_3, and 176 for Δ in equation 8.

$$v_3 = \frac{286}{176}$$
$$= 1.625 \text{ V}$$

Therefore, the value of the voltage v_3 is $\boxed{1.625 \text{ V}}$.